Process Characterization Master Report

A-Mab Drug Substance — PCMR-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Field	Value
Document ID	PCMR-001
Document class	Process Characterization Master Report
Title	Process Characterization Master Report: A-Mab Drug Substance
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—

Role	Function	Name (synthetic)	Signature	Date
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody dependent cellular cytotoxicity; AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; Cpk, process capability index; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ELISA, enzyme linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; ICH, International Council for Harmonisation; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PCMP, Process Characterization Master Plan; PCMR, Process Characterization Master Report; PCP, Process Characterization Plan; PCR, Process Characterization Report; PD, Process Development; PPQ, process performance qualification; PTP, Process Transfer Plan; qPCR, quantitative polymerase chain reaction; RA, risk assessment; RSM, response surface methodology; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TCID50, 50 % tissue culture infectious dose; TFF, tangential flow filtration; TMP, transmembrane pressure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

This report consolidates the Stage 1 process characterization of the A-Mab drug substance process and states the outcome of the campaign as a whole. It summarizes the per step reports listed in Table 1.1 and cites them. It does not re-derive their studies, and every result below is referenced to the report that produced it.

The drug substance process was characterized end to end on qualified scale-down models, from the production bioreactor through to formulation. The campaign covered 37 process parameters across 8 unit operations, of which 22 were studied multivariately in 236 designed runs. A screening design followed by a response surface design was run at 6 of the unit operations. The remaining 2 were characterized univariately. Neither centrifugation and depth filtration nor the ultrafiltration membrane changes the glycan, charge variant or aggregate content of the pool, and §2.3 gives the mechanism in each case.

All 10 drug substance quality attributes met their acceptance criteria in the commercial scale simulation described in §5. Every one of the 2,000 simulated batches met every criterion at once. The lowest capability index was 1.51 on cumulative MVM clearance, and the next lowest was 1.53 on cumulative XMuLV clearance. Both are viral safety attributes judged against a one sided requirement, and both are therefore the attributes with the least room in the process as characterized.

Cumulative viral clearance is 18.87 log10 for XMuLV against a requirement of 16.7 log10, and 10.03 log10 for MVM against a requirement of 8.6 log10. Three steps carry the claim, and they are the low-pH hold, anion exchange chromatography and small virus retentive filtration. Protein A and cation exchange chromatography also remove virus, and no clearance is claimed for either.

Of the 37 parameters, 1 was classified as a critical process parameter, 20 as well controlled critical process parameters, 10 as key process parameters and 6 as general process parameters. The single critical process parameter is the inactivation pH of Step 6. Its normal operating range spans 25 % of the range over which it was characterized, the smallest fraction of any quality-linked process parameter in the campaign.

17 deviations were recorded across 8 executed studies. 15 were retained with a documented impact assessment. The register carries 1 deviation that invalidated a designed study, which was re-executed in full on requalified load material and is reported from the re-execution alone. §8 gives the register and names the report that investigated each entry.

Stage 2 has not been executed. The capability figures in this report are model based estimates built from qualified scale-down data with parameters varying inside their normal operating ranges. They are predictions of commercial performance and will be confirmed during process performance qualification. No design space edge has been run at commercial scale.

1 Introduction

1.1 Purpose

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody dependent cellular cytotoxicity. The drug substance process is being transferred from the development site (Novacyte Biologics — Cambridge, MA (Development)) to the commercial drug substance facility (Novacyte Biologics — Grafton, WI (Commercial DS)). This report consolidates the Stage 1 characterization campaign that supports that transfer. Its purpose is to state what the campaign established about the process, what the resulting control strategy is, and what remains to be demonstrated at commercial scale.

The report is written for an assessor reading the Biologics License Application, and it answers four questions. It says which parameters affect which quality attributes and over what ranges. It states what the process is capable of delivering at commercial scale. It gives the viral clearance the train provides and names the steps credited with it. It says where each of those claims stops.

1.2 The document package

The characterization campaign is documented in 20 controlled documents. PTP-001 frames the transfer and defines its scope. RA-001 is the pre-characterization risk assessment, and its output is the study scope that every plan inherits. PCMP-001 is the master plan over the campaign. Each unit operation then has a plan and a report, PCP-00N and PCR-00N, where N is the process step. This report consolidates the 8 reports. Table 1.1 lists the package.

Table 1.1: The A-Mab drug substance document package. This report consolidates the 8 PCR-00N reports and inherits its study scope from RA-001.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)

Document ID	Document class	Subject
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)

The dependency between the documents is what makes the package readable in one direction. RA-001 scoped the studies, each PCP-00N planned one of them, each PCR-00N executed and reported it, and this report consolidates the set. Where a number in this report differs in the last digit from the same number in a step report, the step report takes precedence. It reports the analysis, and this report reports a summary of it.

1.3 Regulatory basis

The campaign follows the lifecycle approach to process validation, in which process design is Stage 1 and process performance qualification is Stage 2 (U.S. Food and Drug Administration 2011). The enhanced development approach and the design space concept are taken from ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012). Criticality is treated as a continuum and assigned by risk, following ICH

Q9 (International Council for Harmonisation 2023b). The transfer between sites is managed under the pharmaceutical quality system described in ICH Q10 (International Council for Harmonisation 2008). Viral clearance is evaluated as a modular claim under ICH Q5A (International Council for Harmonisation 2023a). The process model, the quality attribute framework and the risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009).

1.4 What this report does not cover

This report covers the drug substance process from the production bioreactor through ultrafiltration and diafiltration. Drug product manufacture, container closure and stability are outside its scope. Analytical method validation is reported separately under the AMV references listed in §7, and this report cites those references without repeating their content. The Stage 2 protocol and its results are not part of this report, and §9 states what they must still demonstrate.

2 Process description and performance

2.1 The process train

The drug substance process is a fed-batch culture followed by the platform purification train established on the Novacyte humanized IgG1 platform and used for three prior products. The order is Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and formulation by ultrafiltration and diafiltration (CMC Biotech Working Group 2009). Table 2.1 gives each step and its principal role in the control strategy.

Table 2.1: The A-Mab drug substance process train, from the production bioreactor through to formulation, with the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The production bioreactor forms the glycan, charge variant and aggregate attributes, so it is the step at which those attributes can be changed by an operating parameter. The purification steps remove impurities and virus and, under platform conditions, do not separate glycosylation variants (CMC Biotech Working Group 2009). Continuous disk stack centrifugation separates the cells from the culture fluid by density, and depth filtration retains the residual debris. Ultrafiltration and diafiltration concentrates the pool to the drug substance target and exchanges it into the formulation buffer.

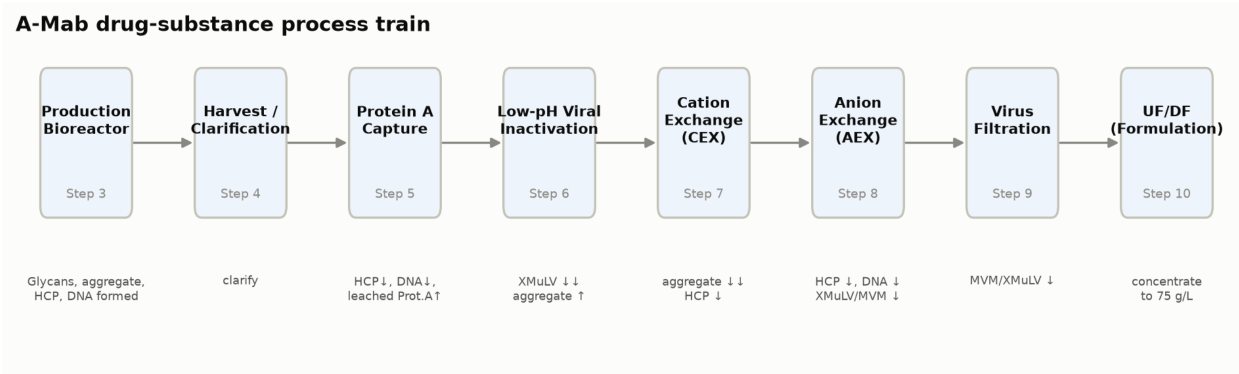

Figure 2.1: The A-Mab drug substance process train with the principal quality impact of each unit operation.

2.2 Scale and performance

The commercial process runs at 15,000 L working volume in the production bioreactor. The nominal harvest titer is 4.37 g/L, and the nominal drug substance concentration after formulation is 75 g/L. A nominal batch delivers 54.6 kg of drug substance at an overall step yield of 83.2 %. Table 2.2 gives the yield of each step and the cumulative yield across the train.

Table 2.2: Nominal step yield, cumulative yield and pool mass across the drug substance train. Step yield is referenced to the production bioreactor harvest.

Step	Unit operation	Step yield (%)	Cumulative yield (%)	Pool mass (g)
3	Production Bioreactor	100.0	100.0	65,596
4	Harvest and Clarification	97.7	97.7	64,071
5	Protein A Chromatography	97.2	95.0	62,293
6	Low-pH Viral Inactivation	99.0	94.0	61,670
7	Cation Exchange Chromatography	95.2	89.5	58,733
8	Anion Exchange Chromatography	97.3	87.1	57,133
9	Small-Virus Retentive Filtration	98.4	85.7	56,232
10	Ultrafiltration / Diafiltration	97.1	83.2	54,588

Cation exchange chromatography carries the largest single step loss in the train, at a step yield of 95.2 %. That loss is a consequence of the step elution used to separate aggregate from monomer, which sacrifices monomer in the trailing edge of the pool in order to reach the aggregate criterion. PCR-007 reports the trade-off in full. No step in the train was operated to maximize yield. Yield carries no acceptance criterion in Table 3.1 and was tracked as a process performance attribute.

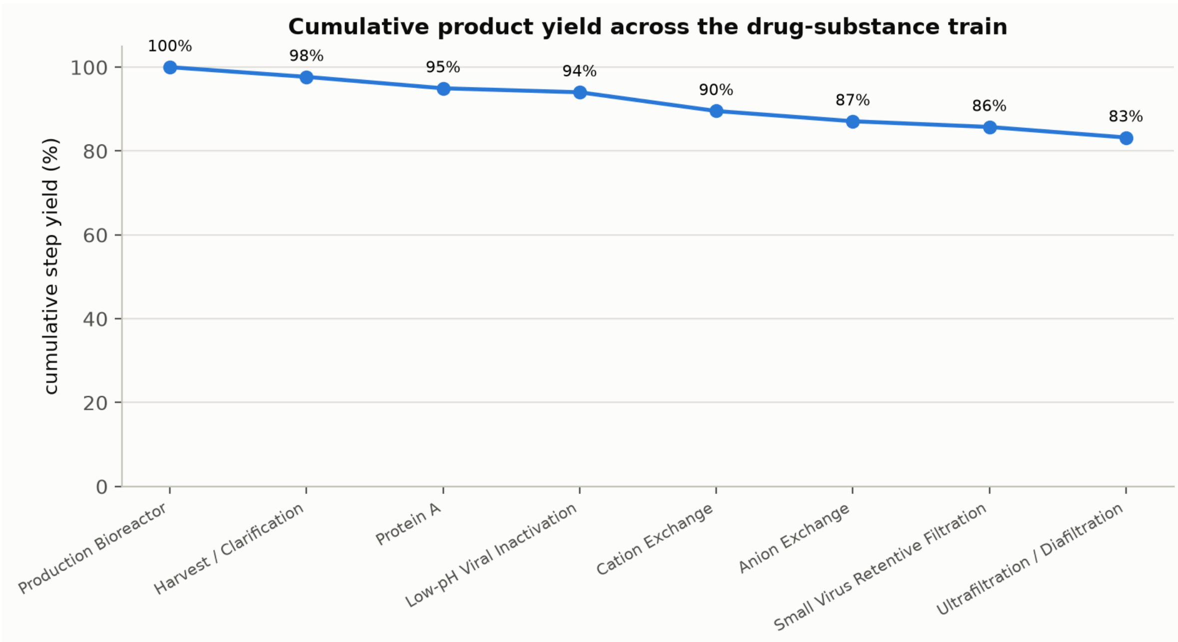

Figure 2.2: Cumulative product step-yield across the eight modelled unit operations.

2.3 Characterization scope

Table 2.3 gives the scope of the campaign by step. It shows how many parameters each step carries, how many were taken into a multivariate study, and how many designed runs the step required. The split between multivariate and univariate study follows the risk ranking in RA-001, in which parameters that scored high for impact on a quality attribute or for interaction with another parameter were taken into the designed experiments and the remainder were bounded univariately.

Table 2.3: Characterization scope by step. DoE runs is the sum of the screening and response surface designs, including centre points.

Step	Unit operation	Parameters	Multivariate	Univariate	DoE runs	Documents
3	Production Bioreactor	9	5	4	47	PCP-003 / PCR-003
4	Harvest and Clarification	3	0	3	0	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	47	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	29	PCP-006 / PCR-006

Step	Unit operation	Parameters	Multivariate	Univariate	DoE runs	Documents
7	Cation Exchange Chromatography	5	4	1	47	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	47	PCP-008 / PCR-008
9	Small-Virus Retentive Filtration	2	2	0	19	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration	3	0	3	0	PCP-010 / PCR-010

The campaign carries no designed experiment at 2 of the steps. Centrifugation and depth filtration act on cells and cell debris, and neither operation binds or elutes the antibody, so the glycan, charge variant and aggregate content of the clarified pool is the content of the harvest. The ultrafiltration membrane retains the antibody and passes water and buffer salts, which raises the concentration of the pool and leaves the molecule unchanged. RA-001 scoped both steps to univariate range confirmation, and PCR-004 and PCR-010 report those studies.

3 Consolidated quality attribute outcomes

3.1 The register

The drug substance quality attribute register holds 10 attributes. It is reproduced as Table 3.1 with the acceptance criterion, the criticality level and the Tool #1 risk score assigned in RA-001. Criticality is not a binary label. It is a level assigned from the severity of the consequence and the uncertainty around the structure and function relationship, following ICH Q9 and the A-Mab framework (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). The attributes at high and very high criticality are the ones the design space is built to protect.

Table 3.1: The A-Mab drug substance quality attribute register, with acceptance criteria, assigned criticality and the Tool #1 risk score from RA-001.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

3 entries in the register sit at very high criticality. High mannose is one of them, and it carries the highest Tool #1 score in the register at 80. The two cumulative viral clearance attributes are the others. Acidic charge variants sit at the other end of the register, at very low criticality and the lowest Tool #1 score at 4. The

attribute is nevertheless carried through the campaign with an upper acceptance limit. Deamidation converts an asparagine residue to aspartate, which adds a negative charge and moves the molecule into the acidic peaks, and the converted fraction rises with the time the pool is held near neutral pH. A 36 hour neutralized hold before anion exchange raised the acidic charge variant content of the load from 21.0 % to 33.5 %, and §8 reports the consequence.

3.2 Where each attribute was studied

Table 3.2 pairs each attribute with the primary step the register assigns it to and with the reports in which the attribute appears as a response of a designed study. The two columns answer different questions. The first is a register assignment. The second lists the steps at which a fitted model predicts the attribute from operating parameters.

Table 3.2: Each drug substance quality attribute, the primary step the register assigns it to, and the reports in which it was carried as a designed-study response.

CQA	Acceptance	Primary step (register)	Characterized as a study response in
Afucosylation	2–13 % of glycans	Production Bioreactor (PCR-003)	PCR-003
Galactosylation (total %Gal)	10–40 % (G1+G2 equiv.)	Production Bioreactor (PCR-003)	PCR-003
High Mannose	3–10 % of glycans	Production Bioreactor (PCR-003)	PCR-003
Aggregates (HMW)	0–5 % HMW (SEC)	Production Bioreactor (PCR-003)	PCR-003, PCR-006, PCR-007
Acidic charge variants (deamidation)	18–40 % (CEX/icIEF)	Production Bioreactor (PCR-003)	PCR-003, PCR-006
Host Cell Protein (HCP)	0–100 ng/mg	Production Bioreactor (PCR-003)	PCR-005, PCR-007, PCR-008
Residual DNA	0–0.001 ng/dose	Production Bioreactor (PCR-003)	no designed study
Leached Protein A	0–5 ppm	Protein A Chromatography (PCR-005)	PCR-005
Viral clearance — XMuLV (enveloped)	16.7–30 log10 (cumulative)	Low-pH Viral Inactivation (PCR-006)	PCR-006, PCR-008, PCR-009
Viral clearance — MVM (parvovirus)	8.6–20 log10 (cumulative)	Anion Exchange Chromatography (PCR-008)	PCR-008, PCR-009

The 3 glycan attributes appear in PCR-003 alone. Afucosylation, galactosylation and high mannose are formed by the culture, and neither the affinity resin nor the two ion exchange resins separates one glycoform from another under platform conditions, so no downstream step carries them as a response (CMC Biotech Working Group 2009). Their control therefore rests on the bioreactor design space alone. The bioreactor design carries 5 responses, more than any other step in the campaign.

5 attributes are carried as a response at more than one step. Aggregate is formed in the bioreactor, raised across the low-pH hold and reduced at cation exchange, and all 3 steps model it. Host cell protein is formed in the culture and is characterized at the 3 chromatography steps that clear it. Cumulative clearance of both model viruses is built from the 3 steps credited in §6. For each of these attributes the drug substance outcome is the product of several steps, and no single report can claim it. §7 states how the in-process criteria at the contributing steps are set so that the cumulative outcome follows.

Residual DNA is carried as a designed response at no step, and Table 3.2 records that. The platform chromatography steps clear DNA by mechanisms that are established from prior products and were not challenged again here (CMC Biotech Working Group 2009). Assurance for residual DNA therefore rests on the capability estimate in §5 and on release testing by the qualified qPCR method, and not on a fitted model. This is the weakest evidential basis of any attribute in the register. The simulated mean sits a factor of 8 below the acceptance limit, and the highest of the 2,000 simulated batches reached 0.00025 ng/dose against a limit of 0.001 ng/dose.

3.3 Outcome

Every attribute in Table 3.1 met its acceptance criterion in the commercial scale simulation, and §5 gives the distribution, the mean and the capability index for each. The two attributes with the least room are the two cumulative viral clearance attributes, and §6 reports how their claims are built. The acceptance criteria in Table 3.1 are the criteria the studies were designed against, and the campaign did not revise any of them.

4 Parameter classification summary

4.1 The consolidated classification

Parameters were classified after the studies were executed, using the decision logic in SOP-4001. A parameter whose variability affects a critical quality attribute is a critical process parameter, and it is a well controlled critical process parameter where the risk of falling outside the design space is low (CMC Biotech Working Group 2009). A key process parameter affects process performance and not product quality. A general process parameter affects neither over the range studied. Table 4.1 gives the counts by step.

Table 4.1: Parameter classification by unit operation across the whole train.

Unit operation	Parameters	CPP	WC-CPP	KPP	GPP
Production Bioreactor	9	0	5	3	1
Harvest / Clarification	3	0	0	2	1
Protein A Chromatography	6	0	2	2	2
Low-pH Viral Inactivation	4	1	2	0	1
Cation Exchange Chromatography	5	0	4	0	1
Anion Exchange Chromatography	5	0	5	0	0
Small Virus Retentive Filtration	2	0	2	0	0
Ultrafiltration / Diafiltration (formulation)	3	0	0	3	0

21 of the 37 parameters are quality-linked process parameters, the critical and the well controlled critical classes together (CMC Biotech Working Group 2009). They are listed with their set-points and normal operating ranges in Table 4.2. These are the parameters the control strategy must hold, and the remaining 16 are monitored for process consistency.

Table 4.2: The quality-linked parameters across the train, with set-point and normal operating range. Each is classified in the report for its step.

Unit operation	Parameter	Unit	Set-point	NOR	Class
Production Bioreactor	Culture pH	pH	6.85	6.75–6.95	WC-CPP
Production Bioreactor	Culture temperature	°C	35	34.5–35.5	WC-CPP

Unit operation	Parameter	Unit	Set-point	NOR	Class
Production Bioreactor	Dissolved CO2 (pCO2)	mmHg	100	70–130	WC-CPP
Production Bioreactor	Osmolality	mOsm	400	375–425	WC-CPP
Production Bioreactor	Culture duration	days	17	16–18	WC-CPP
Protein A Chromatography	Protein load	g/L resin	30	20–40	WC-CPP
Protein A Chromatography	Elution buffer pH	pH	3.55	3.4–3.7	WC-CPP
Low-pH Viral Inactivation	Inactivation pH	pH	3.5	3.4–3.6	CPP
Low-pH Viral Inactivation	Hold time	min	90	60–120	WC-CPP
Low-pH Viral Inactivation	Temperature	°C	21	19–23	WC-CPP
Cation Exchange Chromatography	Protein load	g/L resin	25	18–32	WC-CPP
Cation Exchange Chromatography	Load/Wash conductivity	mS/cm	5	4–6	WC-CPP
Cation Exchange Chromatography	Elution buffer pH	pH	6	5.85–6.15	WC-CPP
Cation Exchange Chromatography	Elution stop collect	OD desc.	1	0.7–1.3	WC-CPP
Anion Exchange Chromatography	Load pH	pH	7.5	7.35–7.65	WC-CPP
Anion Exchange Chromatography	Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	WC-CPP
Anion Exchange Chromatography	Load conductivity	mS/cm	5.5	4–7	WC-CPP
Anion Exchange Chromatography	Protein load	g/L resin	200	120–280	WC-CPP
Anion Exchange Chromatography	Operating flow rate	cm/hr	300	200–400	WC-CPP
Small Virus Retentive Filtration	Filtration volume (load)	L/m2	90	50–105	WC-CPP
Small Virus Retentive Filtration	Filtration pressure	psi	13	10–20	WC-CPP

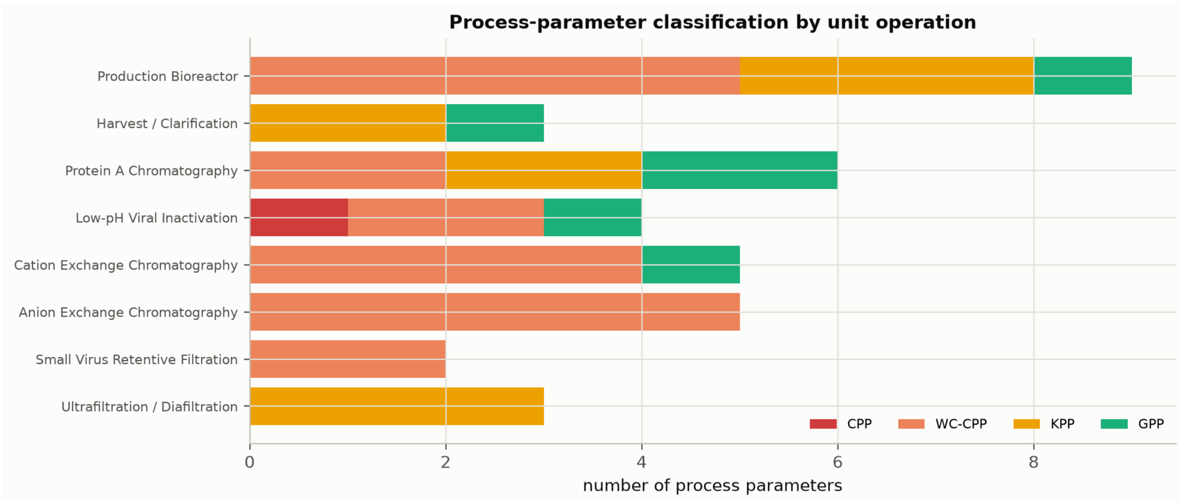

Figure 4.1: Parameter classification by unit operation across the drug substance train.

4.2 The single critical process parameter

Inactivation pH at Step 6 is the only parameter classified as a critical process parameter. It governs the rate of enveloped virus inactivation, and the log reduction achieved by the step falls as the pH rises (CMC Biotech Working Group 2009). Its normal operating range of 3.4 to 3.6 pH units spans 25 % of the range over which it was characterized, against 40 % for the next narrowest quality-linked process parameter. A parameter held in a narrow band relative to the range over which it was studied has a higher risk of leaving the design space on a normal excursion. Inactivation pH was classified as critical on that ground, and PCR-006 records the assessment.

Every other quality-linked process parameter was classified as well controlled. In each case the report for the step recorded the same two grounds. The parameter has a measured effect on a critical quality attribute, and the equipment holds it inside a normal operating range that sits well within the acceptable range proven for it. Each classification was made from the data of its own step.

4.3 Classifications that rest on bounding

Table 4.2 holds 1 quality-linked process parameter whose classification rests on a univariate study rather than on a fitted effect. Operating flow rate at Step 8 was held at its set-point in the designed experiments and varied univariately across 150 to 450 cm/hr. Its classification rests on the absence of an effect over that range and on the control capability of the skid, both reported in PCR-008, and not on a coefficient in a response surface model. A univariate study cannot resolve an interaction between that parameter and another. The classification claims no more than that.

The general process parameters carry the same qualification in a milder form. Each was bounded over a range wider than its normal operating range and showed no effect on any attribute in scope for its step. That is evidence of no effect over the range tested, and it is not evidence that no effect exists outside it. The report for each step gives the range tested in each case.

5 Process capability

5.1 Basis of the estimate

Commercial scale capability was estimated by simulating 2,000 batches of the whole train. In each batch every parameter was drawn from within its normal operating range and the qualified scale-down models were used to propagate the settings to the drug substance attributes. The simulation therefore answers a narrow question. It asks what distribution of drug substance commercial manufacture would produce, given a process held inside the ranges the campaign established and given the effects the campaign measured.

Table 5.1 gives the result for every attribute in the register. The mean and standard deviation are the simulated drug substance distribution, and the capability index is the one sided index against the acceptance limit that applies to that attribute.

Table 5.1: Commercial-scale process capability by quality attribute, from the simulated batches. Cpk is the one-sided index against the acceptance limit that applies to each attribute.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Afucosylation	H	two_sided	2–13	7.98 ± 0.58	2.9
Galactosylation (total %Gal)	H	two_sided	10–40	26 ± 1.6	2.9
High Mannose	VH	two_sided	3–10	5.99 ± 0.51	1.9
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.3
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.1
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2,785.3
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.5
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.5

The simulation produced 0 batches outside an acceptance criterion, from 2,000 simulated. The lowest capability index is 1.51 for cumulative MVM clearance, followed by 1.53 for cumulative XMuLV clearance. The highest is 2,785 for Leached Protein A, where the drug substance acceptance limit sits far above anything the train delivers. The campaign set no acceptance criterion for the capability index, so these values are reported as results and not as a pass or fail against a target.

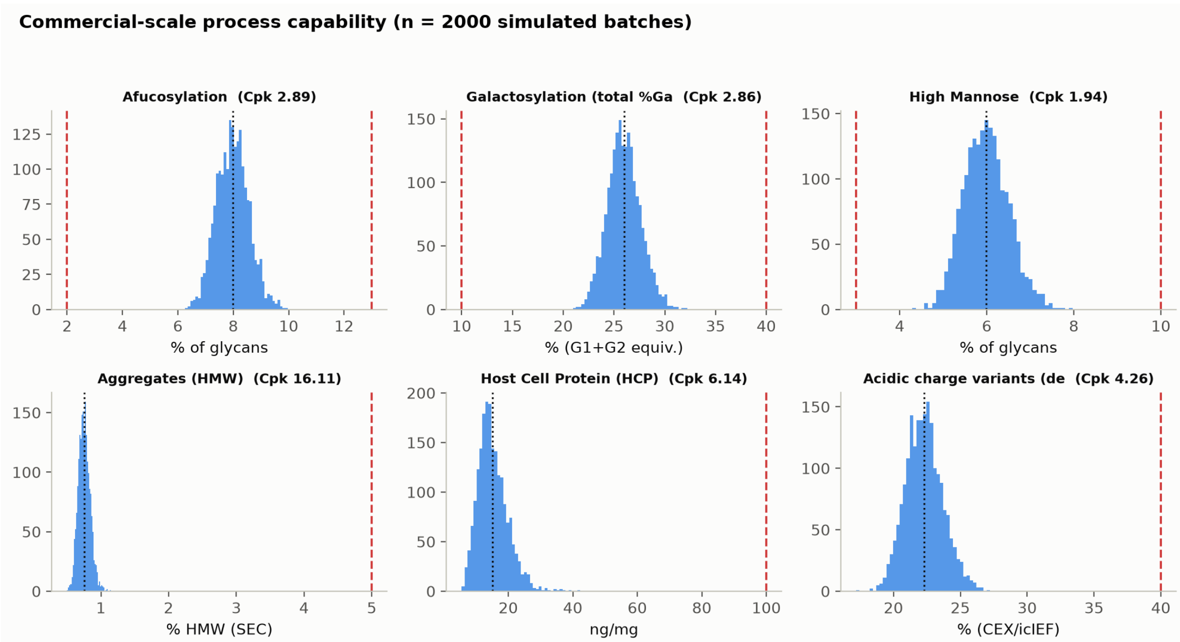

Figure 5.1: Monte-Carlo drug-substance CQA distributions against acceptance limits, annotated with process capability.

5.2 What the estimate is conditioned on

Four conditions bound every capability figure in Table 5.1, and none of them is removed by increasing the size of the simulation.

The estimate is conditioned on the scale-down models being representative. Each model was qualified against development and pilot scale data before its study was run, under SOP-1001, and the report for each step records that qualification.

The estimate is conditioned on the parameters staying inside their normal operating ranges. The simulation samples the normal operating range and not the wider characterized range, so it describes routine operation. It does not describe an excursion, and the proven acceptable ranges in the step reports are the right place to look for the consequence of one.

The estimate is conditioned on the form of the fitted models. A response surface model predicts mean levels over the region it was fitted in (CMC Biotech Working Group 2009). It carries the uncertainty of its own coefficients, and it says nothing about a mechanism that was not in the design.

The estimate is not a commercial scale measurement. No commercial batch has been made. Every figure in Table 5.1 is a prediction to be confirmed during process performance qualification.

6 Viral clearance summary

6.1 The modular claim

Viral safety for A-Mab rests on a modular clearance claim built under ICH Q5A (International Council for Harmonisation 2023a). Each contributing step was spiked and studied independently of the others, and the cumulative claim is the sum of the individual log reduction factors (CMC Biotech Working Group 2009). Two model viruses carry the claim. XMuLV is the model enveloped retrovirus, and MVM is the model small non-enveloped parvovirus and the harder of the two to clear. Table 6.1 gives the contribution of each step and the cumulative total.

Table 6.1: Modular viral clearance by step, and the cumulative claim for each model virus.

Step	XMuLV LRF ($\log_{10}$)	MVM LRF ($\log_{10}$)
Low-pH Viral Inactivation	7.1	0
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.9	10

Cumulative XMuLV clearance is 18.87 log10 against a requirement of 16.7 log10, a margin of 2.17 log10. Cumulative MVM clearance is 10.03 log10 against a requirement of 8.6 log10, a margin of 1.43 log10. MVM therefore carries the smaller margin, and it is also the attribute with the lowest capability index in Table 5.1. The parvovirus claim is built from fewer steps and each of them delivers less.

6.2 Orthogonality

The claim is only additive if the steps remove virus by mechanisms that are independent of one another, so that a virus resistant to one step is not thereby resistant to the next. The 3 credited steps meet that condition. The low-pH hold inactivates by acid denaturation of the viral envelope. Anion exchange holds virus on the positively charged resin at a load pH of 7.5, at which the antibody carries too little negative charge to bind and passes through. Small virus retentive filtration retains virus at the membrane by size and passes the antibody.

The low-pH hold delivers 7.10 log10 against XMuLV and 0 log10 against MVM, because MVM has no envelope to denature. The whole parvovirus claim therefore rests on the 2 steps that remove virus by a physical mechanism, anion exchange at 4.71 log10 and virus filtration at 5.32 log10. Virus filtration is the larger of the two and the more robust, because retention by a membrane does not depend on the load conductivity or pH.

6.3 What is credited and what is not

3 of the 8 unit operations are credited in Table 6.1, and only 2 of them contribute to the parvovirus claim. Protein A chromatography and cation exchange chromatography both remove virus in routine operation on the platform (CMC Biotech Working Group 2009), and no clearance is claimed for either of them here. Excluding them keeps the claim conservative and keeps it independent of two steps whose viral removal was not studied in this campaign. PCR-005 and PCR-007 report those steps without a clearance claim.

§7 explains how the in-process criterion for a viral step is derived from the cumulative requirement minus the nominal contribution of the other credited steps, with an allowance of 0.35 log10 subtracted for the variability of the infectivity assay. The consequence is that no single step is asked to carry the whole requirement, and the failure of any one step to reach its nominal contribution is visible against its own criterion before it reaches the drug substance.

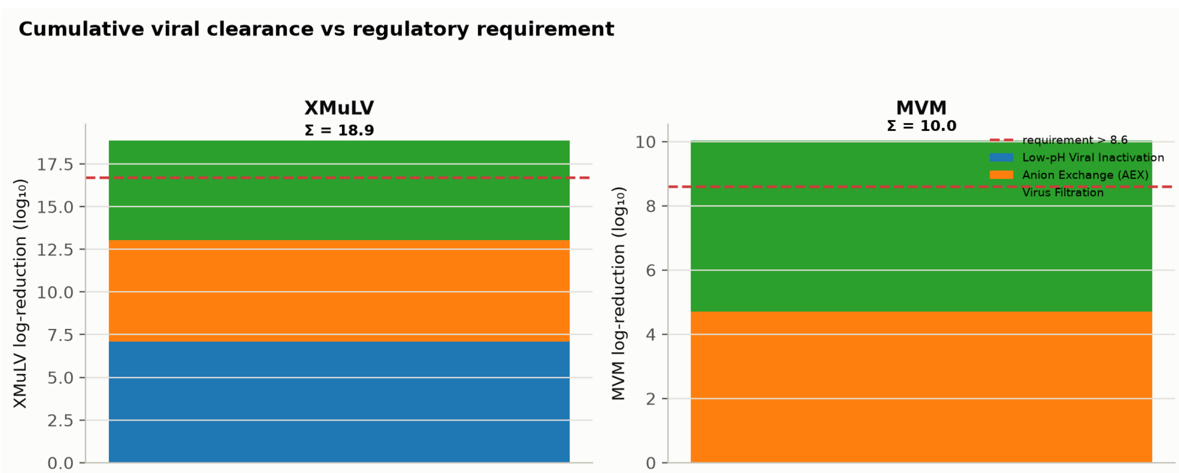

Figure 6.1: Modular viral clearance by step for XMuLV and MVM, against the cumulative requirement.

The log reduction factors are measured on spiked scale-down runs and not on manufacturing material. ICH Q5A sets that out as the basis for a modular claim (International Council for Harmonisation 2023a). The claim is also a nominal claim, in the sense that Table 6.1 evaluates each step at its design centre. Under normal operating range variation the simulated cumulative clearance has a mean of 19.51 log10 for XMuLV and 10.38 log10 for MVM, which sits above the nominal claim in both cases. The nominal claim is therefore the lower of the two figures, and it is the one carried forward into the control strategy.

7 Overall control strategy

7.1 The elements and how they combine

The control strategy has two purposes. It ensures product quality and safety, and it ensures that commercial manufacture is consistent and robust (CMC Biotech Working Group 2009). Product quality is protected by holding every quality-linked process parameter inside a normal operating range that sits within the design space of its step. Process consistency is protected by holding the key process parameters inside their established limits and by monitoring the process attributes that report on them. No single element below carries the strategy alone.

7.2 Parameter ranges and the design space of each step

The first element is the set of ranges in Table 4.2. A normal operating range is where the process is held. The characterized range is the wider region the studies covered. The proven acceptable range for a parameter is the part of the characterized range over which the response still meets its criterion, and each step report gives it per attribute and per parameter.

Table 7.1 states how much of each characterized region meets every criterion that applies to the step. The fitted response surface models were evaluated on an even grid across the coded region of each step, and the table reports the share of grid points that satisfy every criterion at once, together with the response that rejects most of the region.

Table 7.1: Share of each step's characterized region that meets every criterion applying to the step, from the fitted response-surface models evaluated on an even grid.

Step	Unit operation	Grid points	Region meeting every criterion (%)	Limiting response	Rejected by it (%)	Report
3	Production Bioreactor	2401	53.3	Galactosylation	15.5	PCR-003
5	Protein A Chromatography	2401	72.3	Pool HCP (ng/mg)	27.7	PCR-005
6	Low-pH Viral Inactivation	343	80.2	Aggregates (HMW, %)	19.5	PCR-006

Step	Unit operation	Grid points	Region meeting every criterion (%)	Limiting response	Rejected by it (%)	Report
7	Cation Exchange Chromatography	2401	66.6	Aggregates (HMW, %)	22.5	PCR-007
8	Anion Exchange Chromatography	2401	84.5	Pool HCP (ng/mg)	15.5	PCR-008
9	Small-Virus Retentive Filtration	49	73.5	MVM LRF ($\log_{10}$)	26.5	PCR-009

The characterized region is not the design space at any step. At every step some part of the region studied fails a criterion. The Production Bioreactor has the narrowest acceptable region, at 53 % of the region studied. It is also the step with the most criteria to satisfy at once, 5 of them. The limiting response differs by step. Aggregate limits the low-pH hold and cation exchange, and pool host cell protein limits Protein A capture and anion exchange.

The grid in Table 7.1 is evaluated on mean predictions, so a point that just satisfies a criterion on the mean carries roughly even odds of meeting it on an individual batch (CMC Biotech Working Group 2009). The normal operating range of every parameter sits inside its acceptable region. The design space claimed for each step is stated in that step's report and not here.

7.3 In-process controls

The second element is the set of in-process criteria in Table 7.2. An intermediate carries every impurity at the level the steps before it left, and the steps that clear the remainder have not yet run. The Protein A pool carries 9,125 ng/mg host cell protein against a drug substance limit of 100 ng/mg, and it is nevertheless in control, because cation exchange and anion exchange clear it afterwards. The drug substance limit is the wrong yardstick for an intermediate, and the campaign uses a criterion for each step that applies to that step.

Table 7.2: In-process criteria by step, with the basis on which each was set. The "in-process" basis is a criterion derived for the step; "drug substance" means the register criterion applies directly.

Step	Unit operation	In-process measure	Criterion	Basis	Report
3	Production Bioreactor	Afucosylation	2–9.43	in-process	PCR-003
3	Production Bioreactor	Galactosylation	10–30.1	in-process	PCR-003

Step	Unit operation	In-process measure	Criterion	Basis	Report
3	Production Bioreactor	High mannose	3–7.28	in-process	PCR-003
3	Production Bioreactor	Acidic variants	≤ 25.8	in-process	PCR-003
3	Production Bioreactor	Aggregates (HMW)	≤ 1.86	in-process	PCR-003
5	Protein A Chromatography	Pool HCP (ng/mg)	≤ 10,724	in-process	PCR-005
5	Protein A Chromatography	Leached Protein A (ppm)	≤ 5	drug substance	PCR-005
6	Low-pH Viral Inactivation	Aggregates (HMW, %)	≤ 2.17	in-process	PCR-006
6	Low-pH Viral Inactivation	XMuLV LRF ($\log_{10}$)	≥ 4.93	drug substance	PCR-006
6	Low-pH Viral Inactivation	Acidic variants	≤ 40	drug substance	PCR-006
7	Cation Exchange Chromatography	Aggregates (HMW, %)	≤ 1	in-process	PCR-007
7	Cation Exchange Chromatography	Pool HCP (ng/mg)	≤ 193	in-process	PCR-007
8	Anion Exchange Chromatography	Pool HCP (ng/mg)	≤ 21.7	in-process	PCR-008
8	Anion Exchange Chromatography	XMuLV LRF ($\log_{10}$)	≥ 3.78	drug substance	PCR-008
8	Anion Exchange Chromatography	MVM LRF ($\log_{10}$)	≥ 3.28	drug substance	PCR-008
9	Small-Virus Retentive Filtration	MVM LRF ($\log_{10}$)	≥ 4.97	in-process	PCR-009
9	Small-Virus Retentive Filtration	XMuLV LRF ($\log_{10}$)	≥ 5.47	in-process	PCR-009

12 of the 17 criteria are derived for the step and the remaining 5 are the drug substance criterion applied directly. Three methods of derivation are used. For an impurity that the downstream train clears, the drug substance limit is carried back through the clearance the downstream steps deliver and then divided by a safety factor, so that the limit does not depend on every downstream step performing exactly at nominal. For an attribute that no downstream step clears or modifies, such as a glycan, the criterion is set from the demonstrated capability of the process, as an alert limit at 2.5 standard deviations from the simulated mean. For a viral clearance response the criterion is the cumulative requirement minus the nominal contribution of the other credited steps, less the assay allowance described in §6.

The criteria in Table 7.2 are the link between the step reports and the drug substance outcome. Each one is set so that a step meeting its own criterion hands on material the rest of the train can bring to specification. A step that misses its criterion is a deviation at that step and is investigated there, and the consolidated outcome in §3 rests on that.

7.4 Release testing and analytical control

The third element is release testing of the drug substance against the register in Table 3.1, using the validated methods listed in Table 7.3. Release testing is the only element that measures the drug substance itself. It is also the last one, and a result outside a limit at release is found after the batch has been made. Residual DNA is the clearest case. It carries no fitted model at any step, and release testing and the clearance the platform steps deliver are the whole of its assurance.

Table 7.3: Procedures and validated analytical methods cited by the characterization studies. SOP and AMV numbers are placeholders for a synthetic corpus.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP

Reference	Title	Type
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation

The register holds 19 procedures and 10 validated methods. Two procedures apply to every step with a designed experiment, SOP-1001 for the qualification of scale-down models and SOP-1002 for the design and analysis of those experiments. SOP-4001 carries the classification logic used in §4. The step reports cite the subset relevant to each step.

7.5 Equipment and life-cycle controls

The fourth element is the equipment and life-cycle control that keeps the process inside the ranges the studies established. 3 instrumented scale-down systems were under calibration and change control during the campaign, and Table 7.4 records the status of each at the time the studies ran.

Table 7.4: Instrumented scale-down systems under calibration and change control during the campaign.

Equipment	Description	Calibration status	Next calibration due
EQ-CHR-118	temperature-controlled chromatography chamber	in calibration at execution	2026-05-30
EQ-BRX-205	production scale-down bioreactor pCO2 probe	in calibration at execution	2026-06-14
EQ-TFF-142	ultrafiltration / diafiltration TFF skid	in calibration at execution	2026-04-22

All 3 systems were inside their calibration interval when the studies were executed, and all of them appear in the deviation register in §8. Beyond calibration, the life-cycle element covers change control over the process and its controlled documents, resin life-cycle limits under SOP-2008, post-use filter integrity testing under SOP-2012, and continued process verification once commercial manufacture begins.

7.6 Input material controls

The fifth element controls what enters the process. Cell culture media and feed are prepared under SOP-2004 and buffers under SOP-2103, and both carry expiry control. The load material deviation described in §8 raised the acidic charge variant content of the anion exchange load from 21.0 % to 33.5 % and invalidated a designed study. Control of the neutralized hold time between cation exchange and anion exchange is therefore part of the control strategy for Step 8.

7.7 What the control strategy does not cover

The control strategy defined here covers the drug substance process from the production bioreactor to formulation. It does not cover drug product manufacture, container closure or stability. It does not set commercial in-process alert or action limits, which are established from commercial data during Stage 2. It does not cover raw material specifications beyond the preparation and expiry controls named above. Movement within the design space of a step remains subject to the change management system. ICH Q8 does not treat such movement as a regulatory change, and ICH Q10 requires the quality system to assess it (International Council for Harmonisation 2009, 2008).

8 Deviations across the campaign

8.1 The register

17 deviations were recorded across the 8 executed studies. Table 8.1 lists them with the report that investigated each one. 15 were retained, which means the affected data were kept and the report carries a documented impact assessment for them. 1 invalidated a study, and 1 was corrected by modelling and confirmed by verification runs.

Table 8.1: The campaign deviation register. The investigation and the impact assessment for each entry are in the report named in the first column.

Report	Deviation	Summary	Disposition
PCR-003	DEV-003-01	Dissolved-CO2 probe drift during screening run 4	retained
PCR-003	DEV-003-02	Nutrient feed-1 under-delivery on a centre-point batch	retained
PCR-004	DEV-004-01	Depth-filter lot pre-use flush turbidity above alert	retained
PCR-004	DEV-004-02	Post-clarification turbidity excursion above NOR on one run	retained
PCR-005	DEV-005-01	Elution-buffer pH prepared below target on an RSM run	retained
PCR-005	DEV-005-02	Leached Protein A result initially out-of-trend; re-assayed	retained
PCR-006	DEV-006-01	Hold-time record discrepancy between DCS and manual log	retained
PCR-006	DEV-006-02	XMuLV infectivity assay cytotoxicity at low dilution	retained

Report	Deviation	Summary	Disposition
PCR-007	DEV-007-01	Expired Tris load/wash buffer lot used in screening run 6	retained
PCR-007	DEV-007-02	Column inlet temperature excursion during RSM run 14	retained
PCR-008	DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	invalidated and re executed
PCR-008	DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	corrected by modelling and verification runs
PCR-008	DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	retained
PCR-009	DEV-009-01	Transient filtration-pressure excursion above NOR	retained
PCR-009	DEV-009-02	Filter membrane lot flux below vendor-typical value	retained
PCR-010	DEV-010-01	Transmembrane-pressure excursion above NOR during concentration	retained
PCR-010	DEV-010-02	Final DS concentration below target on one run; topped up	retained

8.2 The invalidated study

The first anion exchange design was executed on a non-representative load. An extended neutralized hold of the cation exchange eluate, of 36 hours, promoted asparagine deamidation and raised the acidic charge variant content of the load material to 33.5 %, against 21.0 % in representative material. The deviation was found when the load was verified by icIEF under AMV-3013, after the designs had been run. Both the screening and the response surface designs were invalidated for the purpose of defining an operating region.

The studies were re-executed in full on a requalified, representative load, and PCR-008 reports that execution alone. The first execution is retained as a dataset and is used only to confirm the root cause. It showed an interaction between protein load and wash conductivity that carried the flow-through pool above its in-process host cell protein limit at the corner where both were high, and that interaction is absent from the requalified data. Nothing in the design space or the classification of Step 8 rests on the first execution. The deviation cost a full re-execution of 47 runs.

8.3 The corrected deviation

The trailing edge collection set-point on the anion exchange pool was slightly permissive in both executions, at 180 mAU where 120 mAU was intended. The effect was corrected in the model and then confirmed experimentally in 3 verification runs, whose confidence interval half width was 23.6 on the affected response. PCR-008 gives the correction and the verification.

8.4 The retained deviations

The remaining 15 entries were retained, and they fall into three groups. Instrument and equipment deviations account for the entries against EQ-BRX-205, EQ-CHR-118 and EQ-TFF-142, and each was bounded by the size of the offset against the range the design covered. The dissolved carbon dioxide probe drift at Step 3 is representative. The offset between the in-process probe and the offline reference was 6.5 mmHg, against a characterized range of 40 to 160 mmHg, so the affected run stayed inside the region the design covers. Material and preparation deviations account for the buffer and filter lot entries. The anion exchange equilibration buffer is representative of those. It was released at pH 7.32 against a lower acceptance limit of 7.4 and a target of 7.5. Assay deviations account for the remainder, and each was resolved by re-assay under the qualified method.

PCR-008 carries the most entries at 3. Its study was executed twice. Every entry was dispositioned within the report that recorded it, and Table 8.1 gives that disposition. The impact assessment behind each entry is in the same report, and this report does not repeat it.

9 Conclusions and Stage 2 readiness

9.1 What the campaign established

The A-Mab drug substance process has been characterized across 8 unit operations and 37 process parameters, on qualified scale-down models, in 236 designed runs and the univariate studies that accompany them. The campaign delivered four things.

It delivered a classified parameter set. 21 are quality-linked process parameters and are held within the design space of their step, and the remaining 16 are controlled for process consistency. The single critical process parameter is the inactivation pH of Step 6, for the reason given in §4.

It delivered a design space per step, bounded by the criteria in Table 7.2, together with the proven acceptable ranges reported per attribute and per parameter in the step reports. §7 states how much of each characterized region meets every criterion, and the normal operating ranges sit inside those regions.

It delivered a capability estimate. Every attribute in the register met its acceptance criterion across 2,000 simulated commercial batches, with a lowest capability index of 1.51.

It delivered a modular viral clearance claim of 18.87 log10 for XMuLV and 10.03 log10 for MVM, from 3 steps with independent removal mechanisms, against requirements of 16.7 and 8.6 log10.

9.2 What remains for Stage 2

Stage 2 has not been executed. Process performance qualification must demonstrate at commercial scale what this report predicts from scale-down data. Three things in particular are not yet demonstrated.

Commercial scale performance has not been measured. The capability figures in §5 are model based predictions from simulated batches, and the qualification batches will be the first commercial measurement of the drug substance distribution.

The scale-down models have been qualified but not confirmed against commercial material for this process. Each model was qualified before its study under SOP-1001, and the qualification rests on comparison against development and pilot data. Commercial batches are what close that comparison.

The edges of the characterized regions have not been run at commercial scale. Every design space in this package is defined from scale-down data, and no commercial batch has been run at the edge of one (CMC Biotech Working Group 2009). Movement within a design space remains subject to the change management system (International Council for Harmonisation 2008).

9.3 The bound on the claims in this report

Every claim in this report summarizes a claim made in a step report, and it carries the same bounds. The effects are measured over the characterized ranges and not beyond them. The models predict mean levels and not individual batches. The viral clearance figures are from spiked scale-down runs. Residual DNA carries no fitted model at any step. Where this report states a margin, that margin is the distance between a prediction and a limit, and Stage 2 is what turns the prediction into a measurement.

The process as characterized is ready for Stage 2. The parameters that matter are identified and classified, the ranges that hold them are defined and bounded, the criteria that link the steps to the drug substance are set, and the deviations recorded during the campaign are dispositioned and do not affect the conclusions above.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.